Impact of SNPs on genes

Mapping SNPs to genes

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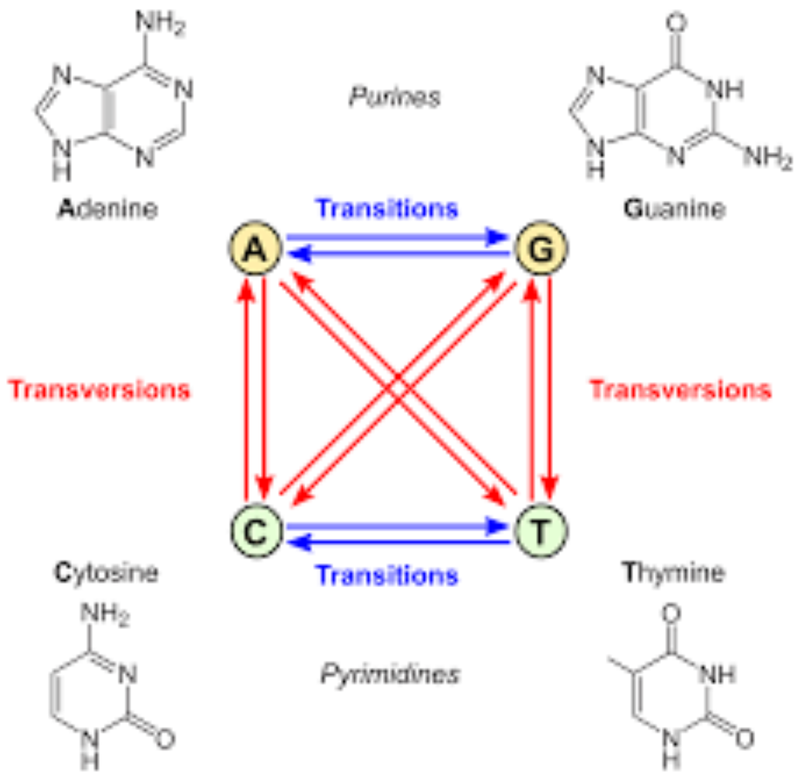
Learning outcomes

- **Software for mapping SNPs to genes**
- **Understand different types of nucleotide changes**
  - Transitions and transversions
- **Understand difference types of mutations**
  - Synonymous and non-synonymous
  - Read-through
  - Stop

Nucleotide changes

Transitions and transversions

- Nucleotide substitutions can take two forms:
  - transitions
    - purine-purine or pyrimidine-pyrimidine interchanges
  - transversions
    - purine-pyrimidine interchanges
- Transitions (G>A|C>T or A>G|T>C) occur more frequently in Ascomycota fungi
- Transition:transversion ratio indicative of certain biological processes



Mutations

In coding regions

- Synonymous (sSNP)
  - A SNP is described as synonymous if there is no change in the AA encoded
    - ‘silent’/evolutionary neutral
- Non-synonymous (nsSNP)
  - Change in the AA encoded
    - sometimes called ‘missense’
- Stop mutation (STP)
  - AA change encodes an AA that is a stop codon, resulting in premature truncation of protein
  - sometimes called ‘nonsense’
- Read-through mutation (RTH)
  - AA change results in stop codon changing to other AA and protein extended

Mutations

In non-coding regions

- Mutations in UTR regions and intergenic regions

Mapping SNPs to genes

snpEff

- Cingolani et al. 2012 PMID: 22728672
- Genomic variant annotations and functions effect prediction
- Has ~38k genomes pre-loaded, or can make custom database for new species/genomes
- Annotate the final vcf
- `java -jar snpEff.jar genome_ref isolate_name.final_SNPs.vcf > isolate_name.snps.annotated`
- Produces an annotated vcf with an additional ‘INFO’ column, and a html report

Mapping SNPs to genes

snpEff

- html results report

Number of effects by functional class

Type (alphabetical order)	Count	Percent
MISSENSE	7,115	34.968%
NONSENSE	46	0.226%
SILENT	13,186	64.806%

Missense / Silent ratio: 0.5396

Number of effects by type and region

Type

Type (alphabetical order)	Count	Percent
downstream_gene_variant	103,013	38.927%
intergenic_region	27,313	10.321%
intron_variant	247	0.093%
missense_variant	7,105	2.685%
non_coding_transcript_exon_variant	1	0%
splice_acceptor_variant	1	0%
splice_donor_variant	1	0%
splice_region_variant	47	0.018%
start_lost	2	0.001%
stop_gained	46	0.017%
stop_lost	8	0.003%
stop_retained_variant	21	0.008%
synonymous_variant	13,165	4.975%
upstream_gene_variant	113,663	42.951%

Region

Type (alphabetical order)	Count	Percent
DOWNSTREAM	103,013	38.934%
EXON	20,325	7.682%
INTERGENIC	27,313	10.323%
INTRON	231	0.087%
SPLICE_SITE_ACCEPTOR	1	0%
SPLICE_SITE_DONOR	1	0%
SPLICE_SITE_REGION	37	0.014%
UPSTREAM	113,663	42.959%

Mapping SNPs to genes

Annotated vcf

- Lots of information
- Perhaps searching for AA that results in drug resistance?
  - Search for gene of interest on FungiDB/NCBI and use ID to search:

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Jo Rhodes@MacBook-Pro-4 Annotated % grep AFUA_4G06890 C101.snps.annotated| grep "missense"
NC_007197.1     1780473 .       T       C       615.04  PASS    AC=1;AF=1.00;AN=1;DP=20;FS=0.000;MLEAC=1;MLEAF=1.00;MQ=60.00;QD=32.37;SOR=1.292;ANN=C|missense_variant|MODERATE|AFUA_4G06890|AFUA_4G06890|transcript|XM_747044.1|protei
n_coding|2/2|c.1279A>G|p.Lys427Glu|1279/1548|1279/1548|427/515||,C|upstream_gene_variant|MODIFIER|AFUA_4G06900|AFUA_4G06900|transcript|XM_747043.1|protein_coding||c.-3022T>C|||||3022| GT:AD:DP:GQ:PL  1:0,19:19:99:625,0
NC_007197.1     1780987 .       C       G       632.04  PASS    AC=1;AF=1.00;AN=1;DP=23;FS=0.000;MLEAC=1;MLEAF=1.00;MQ=60.00;QD=27.48;SOR=1.179;ANN=G|missense_variant|MODERATE|AFUA_4G06890|AFUA_4G06890|transcript|XM_747044.1|protei
n_coding|2/2|c.765G>C|p.Glu255Asp|765/1548|765/1548|255/515||,G|upstream_gene_variant|MODIFIER|AFUA_4G06900|AFUA_4G06900|transcript|XM_747043.1|protein_coding||c.-2508C>G|||||2508|    GT:AD:DP:GQ:PL  1:0,23:23:99:642,0
NC_007197.1     1781009 .       G       T       675.04  PASS    AC=1;AF=1.00;AN=1;DP=23;FS=0.000;MLEAC=1;MLEAF=1.00;MQ=60.00;QD=29.35;SOR=0.963;ANN=T|missense_variant|MODERATE|AFUA_4G06890|AFUA_4G06890|transcript|XM_747044.1|protei
n_coding|2/2|c.743C>A|p.Thr248Asn|743/1548|743/1548|248/515||,T|upstream_gene_variant|MODIFIER|AFUA_4G06900|AFUA_4G06900|transcript|XM_747043.1|protein_coding||c.-2486G>T|||||2486|    GT:AD:DP:GQ:PL  1:0,23:23:99:685,0
NC_007197.1     1781238 .       C       T       689.04  PASS    AC=1;AF=1.00;AN=1;DP=26;FS=0.000;MLEAC=1;MLEAF=1.00;MQ=60.00;QD=29.96;SOR=0.963;ANN=T|missense_variant|MODERATE|AFUA_4G06890|AFUA_4G06890|transcript|XM_747044.1|protei
n_coding|2/2|c.514G>A|p.Val172Met|514/1548|514/1548|172/515||,T|upstream_gene_variant|MODIFIER|AFUA_4G06900|AFUA_4G06900|transcript|XM_747043.1|protein_coding||c.-2257C>T|||||2257|    GT:AD:DP:GQ:PL  1:0,23:23:99:699,0
NC_007197.1     1781459 .       A       T       966.04  PASS    AC=1;AF=1.00;AN=1;DP=31;FS=0.000;MLEAC=1;MLEAF=1.00;MQ=60.00;QD=32.20;SOR=0.693;ANN=T|missense_variant|MODERATE|AFUA_4G06890|AFUA_4G06890|transcript|XM_747044.1|protei
n_coding|2/2|c.293T>A|p.Leu98His|293/1548|293/1548|98/515||,T|upstream_gene_variant|MODIFIER|AFUA_4G06900|AFUA_4G06900|transcript|XM_747043.1|protein_coding||c.-2036A>T|||||2036|      GT:AD:DP:GQ:PL  1:0,30:30:99:976,0
NC_007197.1     1781686 .       T       A       884.04  PASS    AC=1;AF=1.00;AN=1;DP=28;FS=0.000;MLEAC=1;MLEAF=1.00;MQ=60.00;QD=31.57;SOR=0.836;ANN=A|missense_variant|MODERATE|AFUA_4G06890|AFUA_4G06890|transcript|XM_747044.1|protei
n_coding|1/2|c.137A>T|p.Tyr46Phe|137/1548|137/1548|46/515||,A|upstream_gene_variant|MODIFIER|AFUA_4G06900|AFUA_4G06900|transcript|XM_747043.1|protein_coding||c.-1809T>A|||||1809|,A|upstream_gene_variant|MODIFIER|AFUA_4G06910|AFUA_4
G06910|transcript|XM_747042.1|protein_coding||c.-4872T>A|||||4872|      GT:AD:DP:GQ:PL  1:0,28:28:99:894,0
NC_007197.1     1784350 .       T       C       553.04  PASS    AC=1;AF=1.00;AN=1;DP=18;FS=0.000;MLEAC=1;MLEAF=1.00;MQ=60.00;QD=32.53;SOR=0.804;ANN=C|missense_variant|MODERATE|AFUA_4G06900|AFUA_4G06900|transcript|XM_747043.1|protei
n_coding|6/7|c.566T>C|p.Ile189Thr|566/1920|566/1920|189/639||,C|upstream_gene_variant|MODIFIER|AFUA_4G06890|AFUA_4G06890|transcript|XM_747044.1|protein_coding||c.-2528A>G|||||2528|,C|upstream_gene_variant|MODIFIER|AFUA_4G06910|AFUA
_4G06910|transcript|XM_747042.1|protein_coding||c.-2208T>C|||||2208|,C|downstream_gene_variant|MODIFIER|AFUA_4G06920|AFUA_4G06920|transcript|XM_747041.1|protein_coding||c.*4075A>G||||4075|   GT:AD:DP:GQ:PL  1:0,17:17:99:563,0
NC_007197.1     1784694 .       A       G       945.04  PASS    AC=1;AF=1.00;AN=1;DP=28;FS=0.000;MLEAC=1;MLEAF=1.00;MQ=60.00;QD=33.75;SOR=1.382;ANN=G|missense_variant|MODERATE|AFUA_4G06900|AFUA_4G06900|transcript|XM_747043.1|protei
n_coding|6/7|c.910A>G|p.Thr304Ala|910/1920|910/1920|304/639||,G|upstream_gene_variant|MODIFIER|AFUA_4G06890|AFUA_4G06890|transcript|XM_747044.1|protein_coding||c.-2872T>C|||||2872|,G|upstream_gene_variant|MODIFIER|AFUA_4G06910|AFUA
_4G06910|transcript|XM_747042.1|protein_coding||c.-1864A>G|||||1864|,G|downstream_gene_variant|MODIFIER|AFUA_4G06920|AFUA_4G06920|transcript|XM_747041.1|protein_coding||c.*3731T>C|||||3731|   GT:AD:DP:GQ:PL  1:0,28:28:99:955,0
NC_007197.1     1785669 .       T       C       1385.04 PASS    AC=1;AF=1.00;AN=1;DP=31;FS=0.000;MLEAC=1;MLEAF=1.00;MQ=60.00;QD=29.60;SOR=0.756;ANN=C|missense_variant|MODERATE|AFUA_4G06900|AFUA_4G06900|transcript|XM_747043.1|protei
n_coding|7/7|c.1826T>C|p.Leu609Pro|1826/1920|1826/1920|609/639||,C|upstream_gene_variant|MODIFIER|AFUA_4G06890|AFUA_4G06890|transcript|XM_747044.1|protein_coding||c.-3847A>G|||||3847|,C|upstream_gene_variant|MODIFIER|AFUA_4G06910|A
FUA_4G06910|transcript|XM_747042.1|protein_coding||c.-889T>C|||||889|,C|upstream_gene_variant|MODIFIER|AFUA_4G06930|AFUA_4G06930|transcript|XM_747040.1|protein_coding||c.-4457T>C|||||4457|,C|downstream_gene_variant|MODIFIER|AFUA_4G
06920|AFUA_4G06920|transcript|XM_747041.1|protein_coding||c.*2756A>G||||2756|  GT:AD:DP:GQ:PL  1:0,31:31:99:1395,0
Jo Rhodes@MacBook-Pro-4 Annotated %
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